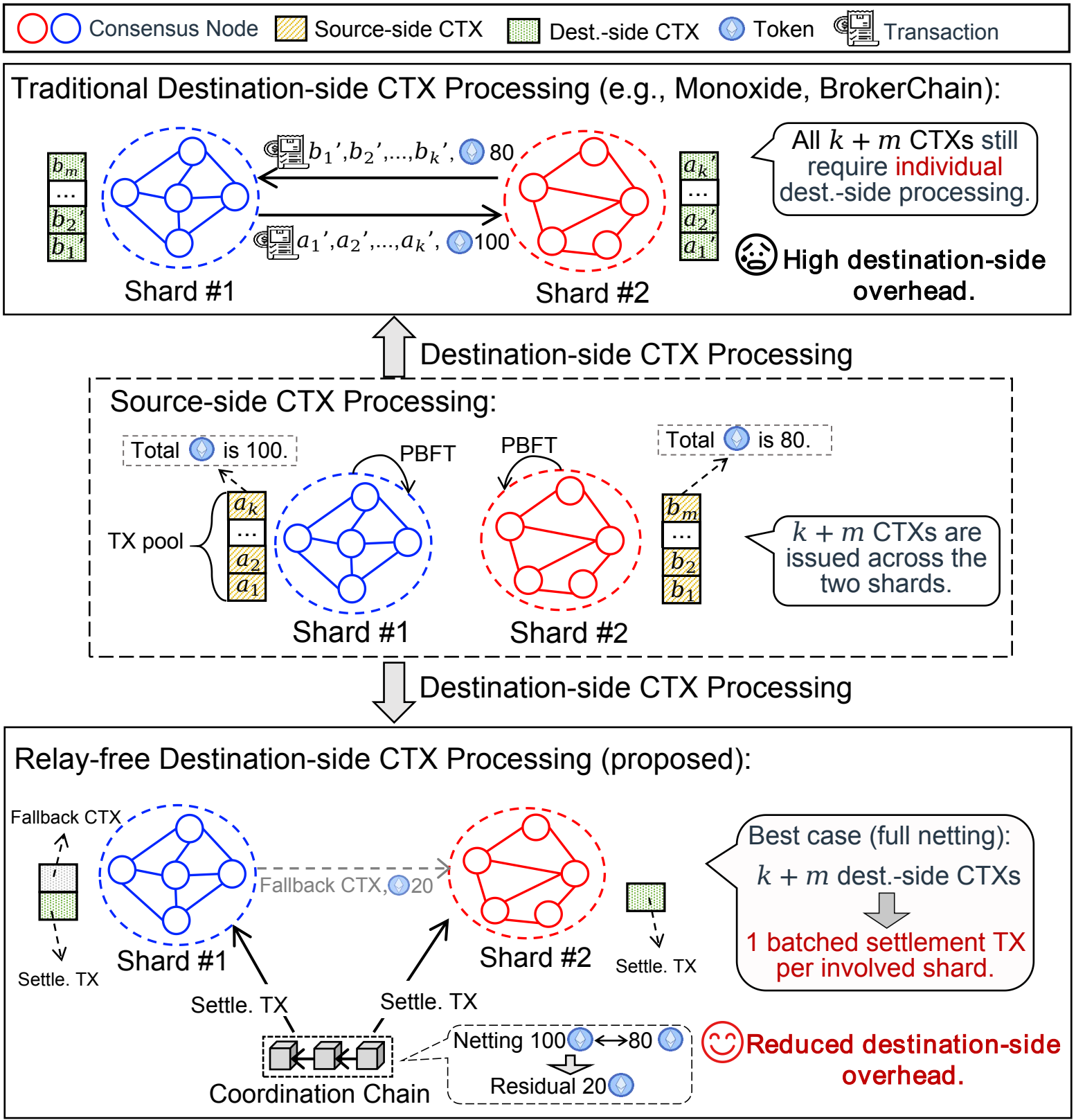

Consensus Node
Source-side CTX
Dest.-side CTX
Token
Transaction
Traditional Destination-side CTX Processing (e.g., Monoxide, BrokerChain):
$b_1', b_2', ..., b_k'$, 80
$a_1', a_2', ..., a_k'$, 100
Shard #1
Shard #2
All $k + m$ CTXs still require individual dest.-side processing.
High destination-side overhead.
Destination-side CTX Processing
Source-side CTX Processing:
Total is 100.
PBFT
PBFT
Total is 80.
TX pool
Shard #1
Shard #2
$k + m$ CTXs are issued across the two shards.
Destination-side CTX Processing
Relay-free Destination-side CTX Processing (proposed):
Fallback CTX
Fallback CTX, 20
Settle. TX
Settle. TX
Settle. TX
Settle. TX
Shard #1
Shard #2
Coordination Chain
Netting 100 ↔ 80
Residual 20
Best case (full netting):
$k + m$ dest.-side CTXs
1 batched settlement TX per involved shard.
Reduced destination-side overhead.